Can LLMs Determine Political Leaning of Polish News Articles?

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***Abstract*—This study investigates the application of Large Language Models (LLMs) to annotate the political and establishment leanings of opening paragraphs in Polish online news articles covering controversial topics. This task, often time-consuming and costly when performed by humans, holds promise for enhancing media fairness and balance cost-effectively. It is one of first attempts to analyze this problem in an under-resourced language like Polish. Findings reveal the challenge of task's difficulty for LLMs and for human annotators. Accuracy of LLMs varies, with better performance on questions related to political leaning than establishment stance. Notably, text-davinci-002 model achieves the highest scores in wing orientation detection and gpt-3.5-turbo excels in recognizing pro-/anti-government bias. This study highlights LLMs limitations for this demanding task in an under-resourced language like Polish, emphasizing the need for alternative detection approaches.**

***Index Terms*—Large Language Models, political bias, Polish online news, establishment bias**

I. Introduction

Contemporary research within the domain of Natural Language Processing (NLP) has recently been affected with the emergence of Large Language Models (LLMs). They are a new competitor for supervised learning methods – LLMs do not need to use annotations, but they are expensive in terms of computing power usage and time required to perform them. On the other hand, people engaged in the manual annotation of textual corpora ought to be suitably prepared for their task, and furthermore remunerated for performing it. In addition, the latter approach is even more time consuming than in the case of LLMs, as humans are not able to work non-stop.

In our work, we investigate if LLMs are capable of automatic annotation of a task that requires a deeper analysis and understanding of texts in under-represented language like Polish. In this study an example of a demanding task is labeling political leanings in Polish news. A positive result from such an experiment would save a lot of time and money in future research in Polish language, and would encourage researchers to undertake topics that could not be addressed due to limited resources. Namely, based on the given fragment of online news article, we determined its political leaning and whether it expresses the characteristics of pro- or anti-government inclination. Reliable results of such experiments could help in determining a stance, such as left- or right-wing, that each of the news suppliers adopts.

The rest of this paper is organized in the following manner: Section II relates to previous works on political leaning with the use of statistical methods and human annotation. In Section III, the details of utilized dataset are presented, while Section IV delineates the methodologies applied in the empirical part of our work. Section V is dedicated to the analysis of the acquired findings. In Section VI we give examples of our results, models' most common mistakes and we talk about the limitations of our approach. Finally, in Section VII we conclude our experimental results and formulate plans for the future research.

II. Related Works

Manual annotation is a costly, tedious, time-consuming, but vital process in the field of NLP. However, throughout the years there has been a lot of research investigating the feasibility of using systems, including most recent LLMs as annotators instead of human beings. Such experiments focused on the model's capacity to automate the process of labeling and categorizing data. It is worth mentioning semi-supervised learning (SSL), which uses big amount of unlabeled data for learning performance improvement, having a small number of labeled examples [1]. Overcoming poor quality and cost with a huge number of examples has been successfully implemented in bias detection task of news article [2].

It has been proved that LLMs can be an efficient alternative to human annotation in providing effective tutoring strategies, taking into consideration student's state [3]. What is more, GPT-3 was defined as capable of reproducing the viewpoints of sub-populations in the USA and examples were provided in which it could be used for social science research, replacing human beings [4]. However, prompting in under-resourced language such as Slovenian yielded worse results for genre annotation task than in English [5].

One must also keep in mind biases and limitations that go along with the use of LLMs, such as ideological or temporal, difficulty in understanding non-literal language which occurs e.g. in news articles, inconsistency in quality of answers as well as difficulty in capturing human intuition [6]. After

taking the political compass test, ChatGPT's results pointed a pro-environmental, left-libertarian orientation, which supports the argument on ideological bias of the model [7]. Another subjective study showed that the AI software tends to create unfavorable limericks concerning conservatives and favorable ones about liberals [8].

There was an attempt to introduce an automated system for measuring media bias [9]. Authors claimed to be able to produce a data-driven bias classification (left-right bias and establishment bias[1]) that were in good agreement with previous classifications based on human judgement. The human annotations were combined and named "The Media Navigator". The method, relying solely on phrase usage frequencies, classified influential media outlets based on their political stance and their establishment bias. However, its provider is "Swiss Policy Research"[2] website and it is considered to be a pseudo-scientific media research project [10].

As for contemporary research on political leanings of news providers in Poland, there is very little of it, and it is qualitative rather than quantitative – it focuses on single examples and does not undertake analysis on multiple examples at once. Study on tabloidization of political discourse in Poland showed that TV media might have a tendency to invite guests and ask for their opinion on current affairs, depending on their political affiliation, which makes the Polish news providers biased [11]. One study used ChatGPT for 25 common NLP tasks in Polish language, some including not too complex analytical reasoning and the system could solve most of the problems considered quite well, except for emotion recognition [12]. Although none of the tasks focused on political leaning detection, the overall results showed that the more demanding the task (requiring objective reasoning such as word sense identification, grammatical correctness or question answering), the worse ChatGPT and *gpt-4* produced the results.

III. Dataset

This section describes the online news dataset which is used for mining the political leanings.

A. Matching news pairs from PONC

In this experiment we use a subset of Polish Online News Corpus (PONC) [13], covering topics considered to be controversial in Poland. To determine whether the reported news had political and establishment leanings, we use news excerpts that cover the same event, but come from two different sources and compare them. In our case, these are the online news of Polish public broadcaster "TVP Info" and a private provider called "TVN 24". In order to select such pairs, we created the following algorithm:

- Load subset of PONC, covering controversial topics,
- Select publish dates with at least 1 article per source, per topic,
- Take descriptions of the articles (first paragraphs) and use SentenceTransformer[3] model to retrieve embeddings,
- Calculate cosine similarities between all possible pairs of articles of opposite news providers within one date,
- Limit results to pair which hold a cosine similarity between 0.60 and 0.95,
- Filter the dataset with the use of human labor to find truly matching articles.

We decided to focus on analyses of news articles' first paragraphs, which we will refer to as descriptions, as they were revealing enough content to guess the topic in comparison to the articles' titles and their length was short enough to be fully processed by the SentenceTransformer model. For embeddings retrieval, we used "st-polish-paraphrase-from-mpnet"[4] model, which is able to map paragraphs and create an up to 768 dimensional vector and is designed for semantic search tasks such as sentence similarity. Cosine similarity lower threshold was chosen based on the ratio of matching pairs to non-matching pairs of descriptions declining fast below this point. We kept the ratio as high as possible and not lower than 50%. The higher threshold was chosen arbitrarily – descriptions which had a cosine similarity higher than 0.95 were almost identical and came from Polish Press Agency (PAP – Polska Agencja Prasowa[5]), thus the articles with identical descriptions were discarded.

Filtering of truly matching descriptions was performed by one person – female researcher in the field of NLP in her 20s, who is the first author of this paper. The annotation guidelines stated that the descriptions must mention the same event, place and actors. If they mentioned the same topic, e.g. protests against new abortion rules, but they considered two different cities in which aforementioned protests took place, then such pair would be regarded as non-matching one.

As a result, we achieved a set with the following numbers of matching pairs of articles per category, listed in Table I.

TABLE I
Number of matching pairs of article descriptions per controversial topic after filtering

Topic	Number of matching pairs	Number of selected pairs
abortion	44	6
church	10	5
constitution	56	8
elections	3	1
eu	38	6
independence	11	3
lgbt	75	9
refugees	17	9
vaccines	49	3

Having prepared the set of matching pairs, we randomly selected 50 pairs comprising of 100 descriptions in total which were used for annotation. No weights were assigned to the

[1] whether media are in favor or against the government
[2] https://swprs.org
[3] https://huggingface.co/sentence-transformers
[4] https://huggingface.co/sdadas/st-polish-paraphrase-from-mpnet
[5] https://www.pap.pl/

topics and the distribution of articles in the final subset is visible in the last column of Table I.

IV. Methods

The following section describes the methods which were used in our experiment in political bias detection.

Human annotators and LLMs were given the same subset of PONC which consisted of randomly selected 50 matching pairs of news articles descriptions. Each of the 100 article descriptions was displayed in random order and the task was to answer the following questions for every description example:

- *On a scale of 1 to 5, where 1 means "strong left-wing views" and 5 means "strong right-wing views," does this introduction to a web article present left-wing or right-wing views? Give only a digit.*
- *On a scale of 1 to 5, where 1 means "strong pro-government" and 5 means "strong anti-government," is this article pro-government or anti-government? Give only a digit.*

Aforementioned prompts were submitted in Polish language.

A. Utilized LLMs

For the LLM-generated answers, we used OpenAI API[6] and the following models[7] were chosen to perform annotation:

- *gpt-4,*
- *gpt-3.5-turbo,*
- *gpt-3.5-turbo-instruct,*
- *text-davinci-002,*
- *text-davinci-003.*

Annotations were generated between 29th August 2023 and 21 September 2023. We generated two different LLM responses per each model with the following temperature levels: "0.2" ensuring more coherent output and "1" that provided more random output. We later performed inter-coder agreement test to see how similar the LLM-generated outputs were [14].

B. Human Annotation

Three adults – two males and one female – volunteered to perform the annotations of the dataset containing 100 articles. The annotations took place between 20th and 29th August and the original dataset was divided into four separate questionnaires, each consisting of randomly selected 25 articles to be annotated. Such division was supposed to ensure non-exhausting annotation for volunteers and therefore, as accurate results as possible. Additionally, before every annotation session, annotators were reminded with an instruction on how to determine left- and right-wing leaning, as well as anti- and pro-government stances. Such approach was to ensure better quality of annotation as well as and higher level of inter-coder agreement [15].

[6]https://openai.com/product

[7]https://platform.openai.com/docs/models

C. Evaluation Metrics

Inter-coder agreement was calculated to check if the annotation guidelines were clear and whether results of human annotation can be considered reproducible. Due to three annotators and a Likert's scale used in the questionnaires, we used Krippendroff's alpha [16]. Secondly, we calculated accuracy as a measure of correct classification of stances by LLM, considering the human annotation as a gold dataset.

V. Results

In this section we present obtained results of performed experiments.

Figure 1 presents Krippendorff's alphas to show inter-coder agreement results of human and LLM annotations within each controversial topic. This case covers a question regarding left- or right-wing leaning of the analyzed article descriptions.

Human annotators achieved higher inter-coder agreement levels in case of *EU* (the highest agreement level), *Church*, *Constitution* and *Refugees* topics. In most of other cases, either *gpt-4* or *gpt-3.5-turbo* models achieved the highest agreement scores, even higher than scores between human annotators. *Independence* and *Vaccines* topics show that *gpt-4* generated exactly the same results for two different temperature levels.

Figure 2 presents Krippendorff's alphas for anti- or pro-government stance presented in articles descriptions.

In this task LLMs achieved on average higher agreement than human annotators in all categories, with *gpt-4* or *gpt-3.5-turbo* achieving best results. However, in case of *Independence* topic, the metric was below zero for gpt-3.5-turbo-instruct and text-davinci-002. The same situation occurred for text-davinci-003 model on *Refugees* topics – these three outcomes indicate lack of reliability and such results cannot be interpreted.

Table II shows accuracy results obtained by each model per each question as well as overall accuracy for both questions. Best accuracy (0.47) for wing leaning question was achieved by *gpt-35-turbo-instruct*. In case of establishment stance question, *gpt-4* had the highest number of matching scores with human annotations (0.51). On average, *gpt-35-turbo* had the best accuracy for both question equal to 0.46, but *gpt-4* was only slightly worse (0.45).

TABLE II
Accuracy scores of each model

Model	Wing leaning	Establishment stance	Overall accuracy
text-davinci-002	0.44	0.23	0.34
text-davinci-003	0.37	0.28	0.33
gpt-35-turbo	0.45	0.46	**0.46**
gpt-35-turbo-instruct	**0.47**	0.30	0.39
gpt-4	0.39	**0.51**	0.45

Figure 3 and Figure 4 present accuracy results as a percentage of correct annotations by each of the LLMs, where human annotations were treated as a point of reference.

On average, it was harder for the models to correctly answer a question regarding the establishment stance (0.361) than the wing leaning one (0.424). The highest number of correctly

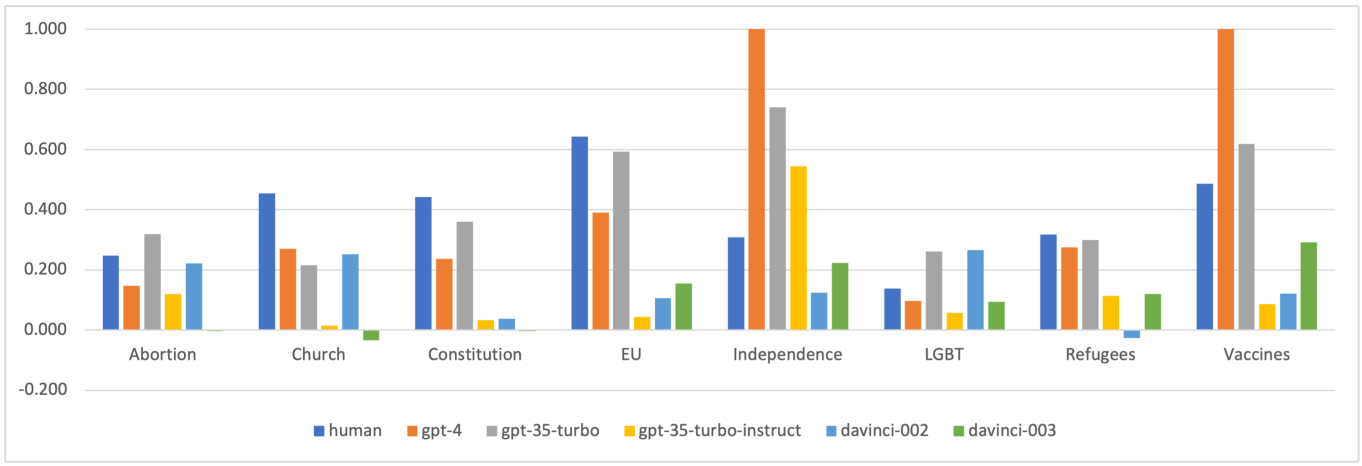


Fig. 1. Inter-coder agreement metric (Krippendorff's alpha) for left- and right-wing leaning per each controversial topic

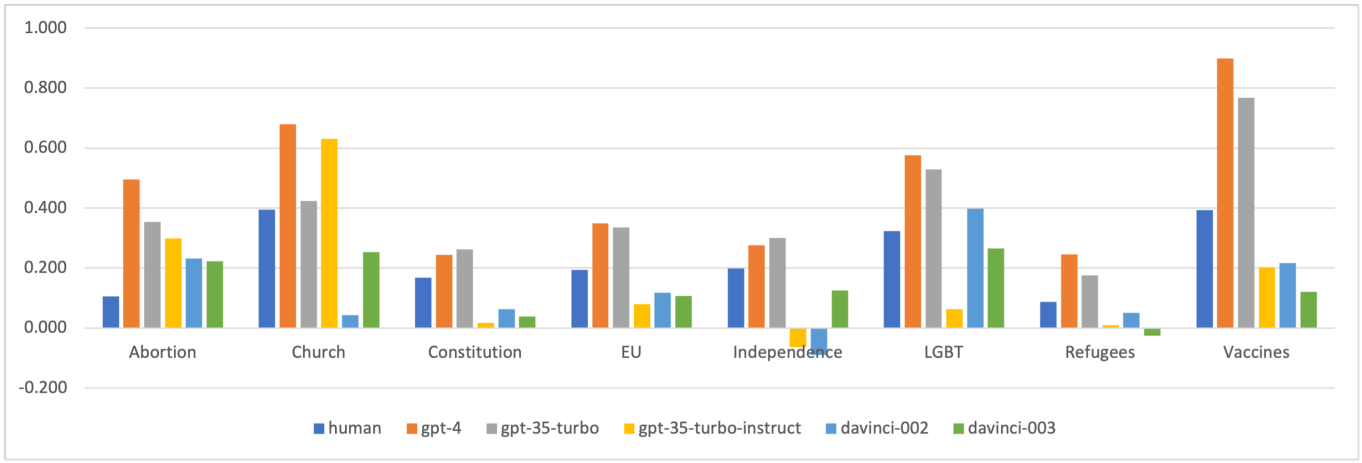


Fig. 2. Inter-coder agreement metric (Krippendorff's alpha) for anti- and pro-government stances per each controversial topic

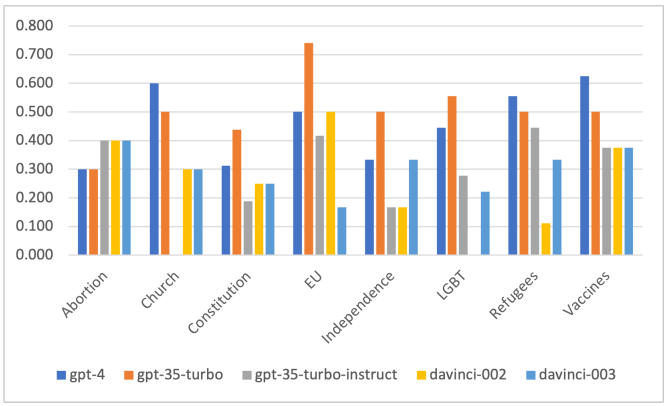


Fig. 3. Accuracy of LLM generated responses to human annotations for anti- and pro-government stances per each controversial topic

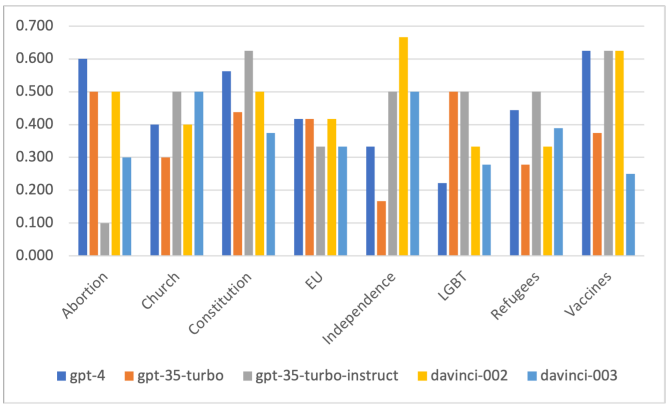


Fig. 4. Accuracy of LLM generated responses to human annotations for left- and right-wing leaning per each controversial topic

answered establishment stance questions was achieved by *gpt-35-turbo* (0.504) and for *EU* topic (0.465), and the lowest one was observed for *text-davinci-002* (0.263) and for *Constitution* topic (0.288). Respectively, the highest accuracy for wing

leaning question was noted for *text-davinci-002* (0.472) and for *Vaccines* and *Constitution* topics (0.500), and the lowest ones for *text-davinci-003* (0.366) and *LGBT* topic (0.367).

VI. Discussion

In this section we describe some mistakes the models made and the ways we handled them as well as limitations of this study.

A. Comparison of human and LLM generated responses

Here, we investigate a couple of examples to hypothesize what could be a cause of matching or completely opposite annotations between people and LLMs.

First example concerns the full agreement on establishment stance between people and *gpt-35-turbo*. The article's description is on EU-related topic and looks as following: *The issuing of orders by First President of the Supreme Court Małgorzata Manowska blocking the full work of the Disciplinary Chamber is an action contrary to Polish law, Justice Minister Zbigniew Ziobro has stated.* The model marked this snippet as "4" (anti-government), just as all human annotators. Possibly this piece of text seems to be easy to process, it is a result of the short length of the text and the fact that it concerns the currently ruling-party. It is worth prompting the same article description once the ruling party in Poland changes and compare the results. However, this approach requires training the model from scratch.

On the other hand, *text-davinci-002* produced completely different results than human annotators which we can interpret as a difficult problem to be handled by this model. The question regarding the government stance concerned the following article: *Przemysław Witkowski, a lecturer at the University of Wrocław and a journalist, was beaten up on one of Wrocław's riverside boulevards. According to him, the attack occurred after he criticised homophobic inscriptions that someone had placed on a wall. The man has a broken nose, cheekbones and zygomatic bones. And the perpetrator is being sought by the police.* All of human annotators gave a score "3" which means they agreed on its neutral overtone. However, the model treated it as strongly anti-government, giving it a score "5". This text is longer than the previous one which could be one of the obstacle for appropriate annotation generation. The other reason for extremely different score is the possible association of critical approach to homophobia of the government which would mean biased input data of the model.

One must keep in mind that the exact mechanics of LLMs is not fully disclosed so it is impossible to explicitly name the reasons behind the discrepancies in answers between people and LLMs.

B. Incorrect output form

Although the prompted message included the request to submit a number as an answer only, LLMs did not always followed this instruction. All the generated responses had to be manually checked and in case of a non-numeric answer changed to a single-figure one. Table III shows numbers of incorrectly generated responses by each model in all iterations. The highest number of such answers can be observed for *text-davinci-002* model (57) and the lowest for *gpt-4* model (11).

TABLE III
Number of non-numeric answers generated by each model, depending on the temperature parameter

	Number of non-numeric answers		
Model	**T = 0.2**	**T = 1**	**Sum**
text-davinci-002	23	34	57
text-davinci-003	21	27	48
gpt-35-turbo	5	39	44
gpt-35-turbo-instruct	22	23	45
gpt-4	6	5	11

Some examples of non-numeric responses included:

- a few additional words indicating that the response refers to a certain score "*Digit: 3 (neutral)", "Level: 1",*
- explanation why certain level was chosen, especially in case of neutral scores: "*This is difficult to define explicitly as the text addresses a variety of issues and cannot be clearly classified as left or right wing. However, if I had to choose a single figure, it would be 3 - neutral/centre.",*
- full sentences: "*The article presents leftist views. I rate it 1."*
- some random characters: "*– 2"*

C. Ambiguous responses

In case of some responses generated by *text-davinci-002* and *text-davinci-003* models, the answers were ambiguous and we introduced the following solutions to handle them:

- two different numbers suggested – choose the non-extreme one "*Depending on the context and individual interpretations, this introduction could be rated as number 4 or 5, i.e. presenting very right-wing views.*" – final answer: 4,
- no number suggested, only text - choose the non-extreme one: "*Moderately anti-government*" - final answer: 2,
- no meaningful answer: "*I give opinions, so I cannot give an answer to the question.*" - final answer: 3.

D. Limitations of this work

Our study is followed by a couple of limitations. Firstly, LLMs are trained on a large amount of text from the Internet and the input content might be biased, resulting in generation of biased responses. They are continuously updated and it is highly probable that in the near future results of a a similar study might be different. Moreover, LLMs are based on statistical methods and unlike people, lack nuance detection skill which might make it harder to detect the stance in text. Additionally, under-resourced languages as Polish, due to limited training data can be harder to process and LLMs can generate less reliable results in them. What is more, in this study we only include article descriptions and not full content – as some generated responses mentioned, it might be not enough to detect the political overtone. Finally, our work analyzed only 100 of article descriptions which can be considered a relatively small number.

VII. Conclusion and Future Work

In this work we showed how LLMs handle annotation of political or establishment leaning of Polish online news articles' opening paragraphs. This task is very costly in time and money when engaging people, hence using LLMs could be beneficial for society by assuring fairer and more balanced media in a timely and cost-effective manner. The results show that not only it is a difficult problem for the LLMs, but it is also a complicated objective to be achieved by human annotators.

LLMs achieved higher agreement than human annotators in all categories, with *gpt-4* or *gpt-3.5-turbo* outperforming other models. However, the metric was below zero for gpt-3.5-turbo-instruct and text-davinci-002 on Independence and Refugees topics, indicating their lack of reliability in these subjects. The best accuracy was achieved for political wing leaning questions, while establishment stance questions were harder for all models. The highest number of correctly answered establishment stance questions was for *gpt-35-turbo* model and *EU* topic, while the lowest accuracy was for *text-davinci-002* model and Constitution topic. For wing leaning question *text-davinci-002* was most accurate (0.472), as well as *Vaccines* and *Constitution* topics (0.500), and the lowest scores were observed for *text-davinci-003* model (0.366) and *LGBT* topic (0.367).

We have demonstrated the limitations of LLMs for such a demanding task and a under-resourced language. We believe that LLMs alone are not enough, therefore new approaches for detectign political leaning in Polish should be investigated in the future.

Our work is just the beginning of efforts towards replacing costly and time-consuming human annotation with LLMs. In the future, we plan to extend the input to full length articles to ensure wider context. Additionally, we think of comparing the annotation agreement rates with the ones of crowd-source workers as our experiments were performed with a small number of annotators. The larger size of a dataset annotated by humans could also influence the results as well as choice of other available approaches e.g. fine-tuned based models. Moreover, we plan to utilize prompts in English and compare the results with instructions in Polish, as under-sourced languages such as Slovenian yielded worse results in the past [5]. Handling more detailed instructions to the LLMs as well as running similar experiments with other language models is advisable.

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